

The selection and use of essential medicines 2025

WHO AWaRe (access, watch, reserve) classification of
antibiotics
for evaluation and monitoring of use



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Suggested citation. The selection and use of essential medicines, 2025: WHO AWaRe (access, watch, reserve) classification of antibiotics for evaluation and monitoring of use. Geneva: World Health Organization; 2025. <https://doi.org/10.2471/B09489>. Licence: [CC BY-NC-SA 3.0 IGO](https://creativecommons.org/licenses/by-nc-sa/3.0/igo).

Cataloguing-in-Publication (CIP) data. CIP data are available at <https://iris.who.int/>.

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Introduction, objectives and methods

Antimicrobial resistance is a threat to global public health and development and is estimated to contribute to more than five million deaths globally each year. The inappropriate use and overuse of antibiotics are driving a global increase in AMR and have a pernicious impact on the effectiveness of these critical medicines. Through the Global Action Plan on AMR, WHO is working to support effective antimicrobial stewardship programmes, improve the surveillance of AMR and reduce inappropriate antibiotic consumption. To assist in the development of tools for antibiotic stewardship at local, national and global levels and to reduce antimicrobial resistance, the AWaRe classification of antibiotics was developed by WHO in which antibiotics are classified into three different groups - Access, Watch and Reserve - based on their clinical importance and the risk of their use promoting resistance to emphasize the importance of their appropriate use.

The AWaRe classification was first introduced in 2017 when antibiotics included in the WHO Model Lists of Essential Medicines were classified as Access, Watch or Reserve by the WHO Expert Committee on Selection and Use of Essential Medicines. In 2019, the Expert Committee recommended that the AWaRe classification should be applied beyond the WHO Model Lists to all commonly used antibiotics globally. Based on advice from the EML Antimicrobials Working Group and recommended by the Expert Committee on Selection and Use of Essential Medicines, all antibiotics in common clinical use globally are now classified using the AWaRe framework. Details of the methodology and evidence underpinning the recommendations can be found in the technical reports of the WHO Expert Committee on Selection and Use of Essential Medicines (WHO Technical Report Series).

The AWaRe classification is intended to be used as a tool for policy makers, researchers and healthcare providers to better support antibiotic prescribing, monitoring and stewardship activities.

WHO Access, Watch, Reserve (AWaRe) classification of antibiotics for evaluation and monitoring of use, 2025

This classification is intended to be used as a tool for countries to better support antibiotic monitoring and stewardship activities. It is not intended as model for the inclusion of antibiotics on national essential medicine lists. Antibiotics classified under AWaRe and also included on the WHO Model Lists of Essential Medicines are indicated in the worksheets.

Antibiotic	Class	ATC code	Category	Listed on EML/EmLc 2025	Notes
Amikacin	Aminoglycosides	J01GB06	Access	Yes	
Amoxicillin	Penicillins	J01CA04	Access	Yes	
Amoxicillin/clavulanic-acid	Beta-lactam/beta-lactamase-inhibitor	J01CR02	Access	Yes	
Amoxicillin/sulbactam	Beta-lactam/beta-lactamase-inhibitor	J01CR02	Access	No	
Ampicillin	Penicillins	J01CA01	Access	Yes	
Ampicillin/sulbactam	Beta-lactam/beta-lactamase-inhibitor	J01CR01	Access	No	
Arbekacin	Aminoglycosides	J01GB12	Watch	No	
Aspoxicillin	Penicillins	J01CA19	Watch	No	
Azandazole	Imidazoles	P01AB04	Access	No	
Azidocillin	Penicillins	J01CE04	Access	No	
Azithromycin	Macrolides	J01FA10	Watch	Yes	
Azocillin	Penicillins	J01CA09	Watch	No	
Aztreonam	Monobactams	J01DF01	Reserve	No	
Aztreonam/avibactam	Monobactam/beta-lactamase-inhibitor	J01DF51	Reserve	No	
Bacampicillin	Penicillins	J01CA08	Access	No	
Bekanamycin	Aminoglycosides	J01GB13	Watch	No	
Benzathine-benzylpenicillin	Penicillins	J01CE08	Access	Yes	
Benzathine-phenoxyethylpenicillin	Penicillins	J01CE10	Access	No	
Benzylpenicillin	Penicillins	J01CE01	Access	Yes	
Blapenem	Carbapenems	J01DH05	Watch	No	
Brodinoprim	Trimethoprim-derivatives	J01EA02	Access	No	
Carbenicillin	Penicillins	J01CA03	Watch	No	
Carindacillin	Penicillins	J01CA05	Watch	No	
Carumonam	Monobactams	J01DF02	Reserve	No	
Cefacetrile	First-generation-cephalosporins	J01DB10	Access	No	
Cefaclor	Second-generation-cephalosporins	J01DC04	Watch	No	
Cefadroxil	First-generation-cephalosporins	J01DB05	Access	No	
Cefalexin	First-generation-cephalosporins	J01DB01	Access	Yes	
Cefaloridine	First-generation-cephalosporins	J01DB02	Access	No	
Cefalotin	First-generation-cephalosporins	J01DB03	Access	No	
Cefamandole	Second-generation-cephalosporins	J01DC03	Watch	No	
Cefapirin	First-generation-cephalosporins	J01DB08	Access	No	
Cefatrizine	First-generation-cephalosporins	J01DB07	Access	No	
Cefazedone	First-generation-cephalosporins	J01DB06	Access	No	
Cefazolin	First-generation-cephalosporins	J01DB04	Access	Yes	
Ceftuperozone	Second-generation-cephalosporins	J01DC13	Watch	No	
Cefcapene-pivoxil	Third-generation-cephalosporins	J01DD17	Watch	No	
Cefdinir	Third-generation-cephalosporins	J01DD15	Watch	No	
Cefditoren-pivoxil	Third-generation-cephalosporins	J01DD16	Watch	No	
Cefepime	Fourth-generation-cephalosporins	J01DE01	Watch	No	
Cefepime-pivoxil	Third-generation-cephalosporins	J01DD10	Watch	No	
Cefiderocol	Other-cephalosporins	J01DI04	Reserve	Yes	
Cefixime	Third-generation-cephalosporins	J01DD08	Watch	Yes	
Cefmenoxime	Third-generation-cephalosporins	J01DD05	Watch	No	
Cefmetazole	Second-generation-cephalosporins	J01DC09	Watch	No	
Cefminox	Second-generation-cephalosporins	J01DC12	Watch	No	
Cefodizime	Third-generation-cephalosporins	J01DD09	Watch	No	
Cefonicid	Second-generation-cephalosporins	J01DC06	Watch	No	
Cefoperazone	Third-generation-cephalosporins	J01DD12	Watch	No	
Ceforanide	Second-generation-cephalosporins	J01DC11	Watch	No	
Cefoselis	Fourth-generation-cephalosporins	not assigned	Watch	No	
Cefotaxime	Third-generation-cephalosporins	J01DD01	Watch	Yes	
Cefotetan	Second-generation-cephalosporins	J01DC05	Watch	No	
Cefotiam	Second-generation-cephalosporins	J01DC07	Watch	No	
Cefoxitin	Second-generation-cephalosporins	J01DC01	Watch	No	
Cefozopran	Fourth-generation-cephalosporins	J01DE03	Watch	No	
Cefpiramide	Third-generation-cephalosporins	J01DD11	Watch	No	
Cefpirome	Fourth-generation-cephalosporins	J01DE02	Watch	No	
Cefpodoxime-proxetil	Third-generation-cephalosporins	J01DD13	Watch	No	
Cefprozil	Second-generation-cephalosporins	J01DC10	Watch	No	
Cefradine	First-generation-cephalosporins	J01DB09	Access	No	
Cefroxadine	First-generation-cephalosporins	J01DB11	Access	No	
Cefsulodin	Third-generation-cephalosporins	J01DD03	Watch	No	
Ceftaroline-fosamil	Fifth-generation-cephalosporins	J01DI02	Reserve	No	
Ceftazidime	Third-generation-cephalosporins	J01DD02	Watch	Yes	
Ceftazidime/avibactam	Third-generation-cephalosporins	J01DD52	Reserve	Yes	
Ceftetam-pivoxil	Third-generation-cephalosporins	J01DD18	Watch	No	
Ceftiole	First-generation-cephalosporins	J01DB12	Access	No	
Ceftibuten	Third-generation-cephalosporins	J01DD14	Watch	No	
Ceftizoxime	Third-generation-cephalosporins	J01DD07	Watch	No	
Ceftobiprole-medocartil	Fifth-generation-cephalosporins	J01DI01	Reserve	No	
Ceftolozane/tazobactam	Fifth-generation-cephalosporins	J01DI54	Reserve	Yes	
Ceftriaxone	Third-generation-cephalosporins	J01DD04	Watch	Yes	
Cefuroxime	Second-generation-cephalosporins	J01DC02	Watch	Yes	
Chloramphenicol	Amphenicols	J01BA01	Access	Yes	
Chlortetracycline	Tetracyclines	J01AA03	Watch	No	
Cinoxacin	Quinolones	J01MB06	Watch	No	
Ciprofloxacin	Fluoroquinolones	J01MA02	Watch	Yes	
Clarithromycin	Macrolides	J01FA09	Watch	Yes	
Clindamycin	Lincosamides	J01FF01	Access	Yes	
Clofodol	Phenol-derivatives	J01XX03	Watch	No	
Clometacin	Penicillins	J01CE07	Access	No	
Clomoxycine	Tetracyclines	J01AA11	Watch	No	
Cloxacillin	Penicillins	J01CF02	Access	Yes	
Colistin_IV	Polymyxins	J01XB01	Reserve	Yes	
Colistin_oral	Polymyxins	A07AA10	Reserve	No	
Dalbavancin	Glycopeptides	J01XA04	Reserve	No	
Dalofopristin/quinupristin	Streptogramins	J01F502	Reserve	No	
Daptomycin	Lipopeptides	J01XX09	Reserve	No	
Delafloxacin	Fluoroquinolones	J01MA23	Watch	No	
Demeclocycline	Tetracyclines	J01AA01	Watch	No	
Dibekacin	Aminoglycosides	J01GB09	Watch	No	
Dicloxacillin	Penicillins	J01CF01	Access	Yes	*therapeutic alternative to cloxacillin
Dinithromycin	Macrolides	J01FA13	Watch	No	
Doripenem	Carbapenems	J01DH04	Watch	No	
Doxycycline	Tetracyclines	J01AA02	Access	Yes	
Enoxacin	Fluoroquinolones	J01MA04	Watch	No	
Epilcillin	Penicillins	J01CA07	Access	No	
Eravacycline	Tetracyclines	J01AA13	Reserve	No	
Ertapenem	Carbapenems	J01DH03	Watch	No	
Erythromycin	Macrolides	J01FA01	Watch	Yes	
Faropenem	Penems	J01DI03	Reserve	No	*therapeutic alternative to clarithromycin (EMLc)
Fidaxomicin	Macrolides	A07AA12	Watch	No	
Fleroxacin	Fluoroquinolones	J01MA08	Watch	No	
Flomoxef	Second-generation-cephalosporins	J01DC14	Watch	No	
Flucloxacillin	Penicillins	J01CF05	Access	Yes	*therapeutic alternative to cloxacillin
Flumequine	Quinolones	J01MB07	Watch	No	
Flurithromycin	Macrolides	J01FA14	Watch	No	
Fosfomycin_IV	Phosphonics	J01XX01	Reserve	Yes	
Fosfomycin_oral	Phosphonics	J01XX01	Watch	No	
Furazidin	Nitrofurant-derivatives	J01XE03	Access	No	
Fusidic-acid	Steroid-antibacterials	J01XC01	Watch	No	
Garenoxacin	Fluoroquinolones	J01MA19	Watch	No	
Gatifloxacin	Fluoroquinolones	J01MA16	Watch	No	
Gemifloxacin	Fluoroquinolones	J01MA15	Watch	No	
Gentamicin	Aminoglycosides	J01GB03	Access	Yes	
Gepotidacin	Triazasonaphthylenes	J01XX13	Watch	No	
Grepafloxacin	Fluoroquinolones	J01MA11	Watch	No	
Hetacillin	Penicillins	J01CA18	Access	No	
Idaprim	Trimethoprim-derivatives	J01EA03	Reserve	No	
Imipenem/cilastatin	Carbapenems	J01DH51	Watch	Yes	
Imipenem/cilastatin/relebactam	Carbapenems	J01DH56	Reserve	No	*therapeutic alternative to meropenem
Isepamicin	Aminoglycosides	J01GB11	Watch	No	
Josamycin	Macrolides	J01FA07	Watch	No	
Kanamycin_IV	Aminoglycosides	J01GB04	Watch	No	
Kanamycin_oral	Aminoglycosides	A07AA08	Watch	No	
Lasclifloxacin	Fluoroquinolones	J01MA25	Watch	No	
Latamoxef	Third-generation-cephalosporins	J01DD06	Watch	No	
Lefamulin	Pleuromutins	J01XX12	Reserve	No	
Levofloxacin	Fluoroquinolones	J01MA12	Watch	No	
Levonadifloxacin	Fluoroquinolones	J01MA24	Watch	No	
Lincomycin	Lincosamides	J01FF02	Watch	No	

Linezolid	Oxazolidinones	J01XX08	Reserve	Yes	
Lomefloxacin	Fluoroquinolones	J01MA07	Watch	No	
Loracarbef	Second-generation-cephalosporins	J01DC08	Watch	No	
Lymscycline	Tetracyclines	J01AA04	Watch	No	
Mecillinam	Penicillins	J01CA11	Access	No	
Meropenem	Carbapenems	J01DH02	Watch	Yes	
Meropenem/vaborbactam	Carbapenems	J01DH52	Reserve	Yes	
Metacycline	Tetracyclines	J01AA05	Watch	No	
Metampicillin	Penicillins	J01CA14	Access	No	
Mefloillin	Penicillins	J01CF03	Access	Yes	*therapeutic alternative to cloxacillin
Metronidazole_IV	Imidazoles	J01XD01	Access	Yes	
Metronidazole_oral	Imidazoles	P01AB01	Access	Yes	
Mezlocillin	Penicillins	J01CA10	Watch	No	
Miconomicin	Aminoglycosides	not assigned	Watch	No	
Midecamycin	Macrolides	J01FA03	Watch	No	
Minocycline_IV	Tetracyclines	J01AA08	Reserve	No	
Minocycline_oral	Tetracyclines	J01AA08	Watch	No	
Miocamycin	Macrolides	J01FA11	Watch	No	
Moxifloxacin	Fluoroquinolones	J01MA14	Watch	No	
Nafcilin	Penicillins	J01CF06	Access	Yes	*therapeutic alternative to cloxacillin
Nalidixic-acid	Quinolones	J01MB02	Watch	No	
Nemonoxacin	Quinolones	J01MB08	Watch	No	
Neomycin_IV	Aminoglycosides	J01GB05	Watch	No	
Neomycin_oral	Aminoglycosides	A07AA01	Watch	No	
Netilmicin	Aminoglycosides	J01GB07	Watch	No	
Nifurtinol	Nitrofuran-derivatives	J01XE02	Access	No	
Nimorazole	Imidazoles	P01AB06	Access	No	
Nitrofurantoin	Nitrofuran-derivatives	J01XE01	Access	Yes	
Norfloxacin	Fluoroquinolones	J01MA06	Watch	No	
Ofloxacin	Fluoroquinolones	J01MA01	Watch	No	
Oleandomycin	Macrolides	J01FA05	Watch	No	
Omadacycline	Tetracyclines	J01AA15	Reserve	No	
Oritavancin	Glycopeptides	J01XA05	Reserve	No	
Ornidazole_IV	Imidazoles	J01XD03	Access	No	
Ornidazole_oral	Imidazoles	P01AB03	Access	No	
Oxacillin	Penicillins	J01CF04	Access	Yes	*therapeutic alternative to cloxacillin
Oxolinic-acid	Quinolones	J01MB05	Watch	No	
Oxytetracycline	Tetracyclines	J01AA06	Watch	No	
Panipenem	Carbapenems	J01DH55	Watch	No	
Paromomycin_oral	Aminoglycosides	A07AA06	Watch	No	
Pazufloxacin	Fluoroquinolones	J01MA18	Watch	No	
Pefloxacin	Fluoroquinolones	J01MA03	Watch	No	
Penamceillin	Penicillins	J01CE06	Access	No	
Penimepicycline	Tetracyclines	J01AA10	Watch	No	
Phenetollin	Penicillins	J01CE05	Watch	No	
Phenoxy-methylpenicillin	Penicillins	J01CE02	Access	Yes	
Pipemidic-acid	Quinolones	J01MB04	Watch	No	
Piperacilin	Penicillins	J01CA12	Watch	No	
Piperacilin/tazobactam	Beta-lactam/beta-lactamase-inhibitor_anti-ps	J01CR05	Watch	Yes	
Piromidic-acid	Quinolones	J01MB03	Watch	No	
Pivampicillin	Penicillins	J01CA02	Access	No	
Pivmecillinam	Penicillins	J01CA08	Access	No	
Piazomicin	Aminoglycosides	J01GB14	Reserve	Yes	
Polymyxin-B_IV	Polymyxins	J01XB02	Reserve	Yes	
Polymyxin-B_oral	Polymyxins	A07AA05	Reserve	No	
Pristinamycin	Streptogramins	J01FG01	Watch	No	
Procaine-benzylpenicillin	Penicillins	J01CE09	Access	Yes	
Progenidazole	Imidazoles	P01AB05	Access	No	
Propicilin	Penicillins	J01CE03	Access	No	
Prulifloxacin	Fluoroquinolones	J01MA17	Watch	No	
Ribostamycin	Aminoglycosides	J01GB10	Watch	No	
Rifabutin	Rifamycins	J04AB04	Watch	No	
Rifampicin	Rifamycins	J04AB02	Watch	No	
Rifamycin_IV	Rifamycins	J04AB03	Watch	No	
Rifamycin_oral	Rifamycins	A07AA13	Watch	No	
Rifaximin	Rifamycins	A07AA11	Watch	No	
Rokitamycin	Macrolides	J01FA12	Watch	No	
Rolitetracycline	Tetracyclines	J01AA09	Watch	No	
Rosoxacin	Quinolones	J01MB01	Watch	No	
Roxithromycin	Macrolides	J01FA06	Watch	No	
Rufloxacin	Fluoroquinolones	J01MA10	Watch	No	
Sarecycline	Tetracyclines	J01AA14	Watch	No	
Satranidazole	Imidazoles	P01AB08	Access	No	
Secnidazole	Imidazoles	P01AB07	Access	No	
Sisomicin	Aminoglycosides	J01GB08	Watch	No	
Sitafloxacin	Fluoroquinolones	J01MA21	Watch	No	
Solithromycin	Macrolides	J01FA16	Watch	No	
Sparfloxacin	Fluoroquinolones	J01MA09	Watch	No	
Spectinomycin	Aminocyclitols	J01XX04	Access	Yes	
Spiramycin	Macrolides	J01FA02	Watch	No	
Streptoduocin	Aminoglycosides	J01GA02	Watch	No	
Streptomycin_IV	Aminoglycosides	J01GA01	Watch	No	
Streptomycin_oral	Aminoglycosides	A07AA04	Watch	No	
Sulbactam	Beta-lactamase-inhibitors	J01CG01	Access	No	
Sulbenicillin	Penicillins	J01CA16	Watch	No	
Sulfadiazine	Sulfonamides	J01EC02	Access	No	
Sulfadiazine/tetroxoprim	Sulfonamide-trimethoprim-combinations	J01EE06	Access	No	
Sulfadiazine/trimethoprim	Sulfonamide-trimethoprim-combinations	J01EE02	Access	No	
Sulfadimethoxine	Sulfonamides	J01ED01	Access	No	
Sulfadimidine	Sulfonamides	J01EB03	Access	No	
Sulfadimidine/trimethoprim	Sulfonamide-trimethoprim-combinations	J01EE05	Access	No	
Sulfafurazole	Sulfonamides	J01EB05	Access	No	
Sulfasizodimidine	Sulfonamides	J01EB01	Access	No	
Sulfalene	Sulfonamides	J01ED02	Access	No	
Sulfamazone	Sulfonamides	J01ED09	Access	No	
Sulfamerazine	Sulfonamides	J01ED07	Access	No	
Sulfamerazine/trimethoprim	Sulfonamide-trimethoprim-combinations	J01EE07	Access	No	
Sulfamethizole	Sulfonamides	J01EB02	Access	No	
Sulfamethoxazole	Sulfonamides	J01EE01	Access	No	
Sulfamethoxazole/trimethoprim	Sulfonamide-trimethoprim-combinations	J01EE01	Access	Yes	
Sulfamethoxypyridazine	Sulfonamides	J01ED05	Access	No	
Sulfametomidine	Sulfonamides	J01ED03	Access	No	
Sulfametoxydiazine	Sulfonamides	J01ED04	Access	No	
Sulfametrole/trimethoprim	Sulfonamide-trimethoprim-combinations	J01EE03	Access	No	
Sulfamoxole	Sulfonamides	J01EC03	Access	No	
Sulfamoxole/trimethoprim	Sulfonamide-trimethoprim-combinations	J01EE04	Access	No	
Sulfanilamide	Sulfonamides	J01EB06	Access	No	
Sulfaperin	Sulfonamides	J01ED06	Access	No	
Sulfaphenazole	Sulfonamides	J01ED08	Access	No	
Sulfapyridine	Sulfonamides	J01EB04	Access	No	
Sulfathiazole	Sulfonamides	J01EB07	Access	No	
Sulfathiourea	Sulfonamides	J01EB08	Access	No	
Sulfamulin	Beta-lactam/beta-lactamase-inhibitor	J01CR04	Access	No	
Talampicillin	Penicillins	J01CA15	Access	No	
Tazobactam	Beta-lactamase-inhibitors	J01CG02	Watch	No	
Tebipenem	Carbapenems	J01DH06	Watch	No	
Tedizolid	Oxazolidinones	J01XX11	Reserve	Yes	*therapeutic alternative to linezolid (EML)
Teicoplanin	Glycopeptides	J01XA02	Watch	No	
Telavancin	Glycopeptides	J01XA03	Reserve	No	
Telithromycin	Macrolides	J01FA15	Watch	No	
Temafloxacin	Fluoroquinolones	J01MA05	Watch	No	
Temocillin	Penicillins	J01CA17	Watch	No	
Tetracycline	Tetracyclines	J01AA07	Access	No	
Thiamphenicol	Amphenicols	J01BA02	Access	No	
Ticarcillin	Penicillins	J01CA13	Watch	No	
Ticarcillin/clavulanic-acid	Beta-lactam/beta-lactamase-inhibitor	J01CR03	Watch	No	
Tigecycline	Glycylcyclines	J01AA12	Reserve	No	
Tinidazole_IV	Imidazoles	J01XD02	Access	No	
Tinidazole_oral	Imidazoles	P01AB02	Access	No	
Tobramycin	Aminoglycosides	J01GB01	Watch	No	
Tosufloxacin	Fluoroquinolones	J01MA22	Watch	No	
Trimethoprim	Trimethoprim-derivatives	J01EA01	Access	Yes	
Troleandomycin	Macrolides	J01FA08	Watch	No	
Trovafloxacin	Fluoroquinolones	J01MA13	Watch	No	
Vancomycin_IV	Glycopeptides	J01XA01	Watch	Yes	
Vancomycin_oral	Glycopeptides	A07AA09	Watch	Yes	

Access group antibiotics

This group includes antibiotics that have activity against a wide range of commonly encountered susceptible pathogens while also showing lower resistance potential than antibiotics in the other groups. Selected Access group antibiotics are recommended as essential first or second choice empiric treatment options for infectious syndromes reviewed by the EML Expert Committee and are listed as individual medicines on the Model Lists of Essential Medicines to improve access and promote appropriate use.

Antibiotic	Class	ATC code	Category	Listed on EML/EMLc 2025	Notes
Amikacin	Aminoglycosides	J01GB06	Access	Yes	0
Amoxicillin	Penicillins	J01CA04	Access	Yes	0
Amoxicillin/clavulanic-acid	Beta-lactam/beta-lactamase-inhibitor	J01CR02	Access	Yes	0
Amoxicillin/sulbactam	Beta-lactam/beta-lactamase-inhibitor	J01CR02	Access	No	0
Ampicillin	Penicillins	J01CA01	Access	Yes	0
Ampicillin/sulbactam	Beta-lactam/beta-lactamase-inhibitor	J01CR01	Access	No	0
Azanidazole	Imidazoles	P01AB04	Access	No	0
Azidocillin	Penicillins	J01CE04	Access	No	0
Bacampicillin	Penicillins	J01CA06	Access	No	0
Benzathine-benzylpenicillin	Penicillins	J01CE08	Access	Yes	0
Benzathine-phenoxymethylpenicillin	Penicillins	J01CE10	Access	No	0
Benzylpenicillin	Penicillins	J01CE01	Access	Yes	0
Brodinoprim	Trimethoprim-derivatives	J01EA02	Access	No	0
Cefacetrile	First-generation-cephalosporins	J01DB10	Access	No	0
Cefadroxil	First-generation-cephalosporins	J01DB05	Access	No	0
Cefalexin	First-generation-cephalosporins	J01DB01	Access	Yes	0
Cefaloridine	First-generation-cephalosporins	J01DB02	Access	No	0
Cefalotin	First-generation-cephalosporins	J01DB03	Access	No	0
Cefapirin	First-generation-cephalosporins	J01DB08	Access	No	0
Cefatrizine	First-generation-cephalosporins	J01DB07	Access	No	0
Cefazedone	First-generation-cephalosporins	J01DB06	Access	No	0
Cefazolin	First-generation-cephalosporins	J01DB04	Access	Yes	0
Cefradine	First-generation-cephalosporins	J01DB09	Access	No	0
Cefroxadine	First-generation-cephalosporins	J01DB11	Access	No	0
Ceftazole	First-generation-cephalosporins	J01DB12	Access	No	0
Chloramphenicol	Amphenicols	J01BA01	Access	Yes	0
Clindamycin	Lincosamides	J01FF01	Access	Yes	0
Clometocillin	Penicillins	J01CE07	Access	No	0
Cloxacillin	Penicillins	J01CF02	Access	Yes	0
Dicloxacillin	Penicillins	J01CF01	Access	Yes	*therapeutic alternative to cloxacillin
Doxycycline	Tetracyclines	J01AA02	Access	Yes	0
Epacillin	Penicillins	J01CA07	Access	No	0
Flucloxacillin	Penicillins	J01CF05	Access	Yes	*therapeutic alternative to cloxacillin
Furazidin	Nitrofurans-derivatives	J01XE03	Access	No	0
Gentamicin	Aminoglycosides	J01GB03	Access	Yes	0
Hetacillin	Penicillins	J01CA18	Access	No	0
Mecillinam	Penicillins	J01CA11	Access	No	0
Metampicillin	Penicillins	J01CA14	Access	No	0
Meticillin	Penicillins	J01CF03	Access	Yes	*therapeutic alternative to cloxacillin
Metronidazole_IV	Imidazoles	J01XD01	Access	Yes	0
Metronidazole_oral	Imidazoles	P01AB01	Access	Yes	0
Nafcillin	Penicillins	J01CF06	Access	Yes	*therapeutic alternative to cloxacillin
Nifurtolol	Nitrofurans-derivatives	J01XE02	Access	No	0
Nimorazole	Imidazoles	P01AB06	Access	No	0
Nitrofurantoin	Nitrofurans-derivatives	J01XE01	Access	Yes	0
Ornidazole_IV	Imidazoles	J01XD03	Access	No	0
Ornidazole_oral	Imidazoles	P01AB03	Access	No	0
Oxacillin	Penicillins	J01CF04	Access	Yes	*therapeutic alternative to cloxacillin
Penamocillin	Penicillins	J01CE06	Access	No	0
Phenoxymethylpenicillin	Penicillins	J01CE02	Access	Yes	0
Pivampicillin	Penicillins	J01CA02	Access	No	0
Pivmecillinam	Penicillins	J01CA08	Access	No	0
Procaine-benzylpenicillin	Penicillins	J01CE09	Access	Yes	0
Propenidazole	Imidazoles	P01AB05	Access	No	0
Propicillin	Penicillins	J01CE03	Access	No	0
Satranidazole	Imidazoles	P01AB08	Access	No	0
Secnidazole	Imidazoles	P01AB07	Access	No	0
Spectinomycin	Aminocyclitols	J01XX04	Access	Yes	0
Sulbactam	Beta-lactamase-inhibitors	J01CG01	Access	No	0
Sulfadiazine	Sulfonamides	J01EC02	Access	No	0
Sulfadiazine/tetroxoprim	Sulfonamide-trimethoprim-combinations	J01EE06	Access	No	0
Sulfadiazine/trimethoprim	Sulfonamide-trimethoprim-combinations	J01EE02	Access	No	0
Sulfadimethoxine	Sulfonamides	J01ED01	Access	No	0
Sulfadimidine	Sulfonamides	J01EB03	Access	No	0
Sulfadimidine/trimethoprim	Sulfonamide-trimethoprim-combinations	J01EE05	Access	No	0
Sulfafurazole	Sulfonamides	J01EB05	Access	No	0
Sulfaisodimidine	Sulfonamides	J01EB01	Access	No	0
Sulfalene	Sulfonamides	J01ED02	Access	No	0
Sulfamazone	Sulfonamides	J01ED09	Access	No	0
Sulfamerazine	Sulfonamides	J01ED07	Access	No	0
Sulfamerazine/trimethoprim	Sulfonamide-trimethoprim-combinations	J01EE07	Access	No	0
Sulfamethizole	Sulfonamides	J01EB02	Access	No	0
Sulfamethoxazole	Sulfonamides	J01EC01	Access	No	0
Sulfamethoxazole/trimethoprim	Sulfonamide-trimethoprim-combinations	J01EE01	Access	Yes	0
Sulfamethoxypyridazine	Sulfonamides	J01ED05	Access	No	0
Sulfametomidine	Sulfonamides	J01ED03	Access	No	0
Sulfamethoxydiazine	Sulfonamides	J01ED04	Access	No	0
Sulfamethole/trimethoprim	Sulfonamide-trimethoprim-combinations	J01EE03	Access	No	0
Sulfamoxole	Sulfonamides	J01EC03	Access	No	0
Sulfamoxole/trimethoprim	Sulfonamide-trimethoprim-combinations	J01EE04	Access	No	0
Sulfanilamide	Sulfonamides	J01EB06	Access	No	0
Sulfaperin	Sulfonamides	J01ED06	Access	No	0
Sulfaphenazole	Sulfonamides	J01ED08	Access	No	0
Sulfapyridine	Sulfonamides	J01EB04	Access	No	0
Sulfathiazole	Sulfonamides	J01EB07	Access	No	0
Sulfathiourea	Sulfonamides	J01EB08	Access	No	0
Sulfamicrocin	Beta-lactam/beta-lactamase-inhibitor	J01CR04	Access	No	0
Talampicillin	Penicillins	J01CA15	Access	No	0
Tetracycline	Tetracyclines	J01AA07	Access	No	0
Thiamphenicol	Amphenicols	J01BA02	Access	No	0
Tinidazole_IV	Imidazoles	J01XD02	Access	No	0
Tinidazole_oral	Imidazoles	P01AB02	Access	No	0
Trimethoprim	Trimethoprim-derivatives	J01EA01	Access	Yes	0

Watch group antibiotics

This group includes antibiotic classes that have higher resistance potential and includes most of the highest priority agents among the Critically Important Antimicrobials for Human Medicine¹ and/or antibiotics that are at relatively high risk of selection of bacterial resistance. These medicines should be prioritized as key targets of stewardship programs and monitoring. Selected Watch group antibiotics are recommended as essential first or second choice empiric treatment options for a limited number of specific infectious syndromes and are listed as individual medicines on the WHO Model Lists of Essential Medicines.

¹Critically Important Antimicrobials for Human Medicine 6th Revision 2018

Antibiotic	Class	ATC code	Category	Listed on EML/EMLc 2025	Notes
Arbekacin	Aminoglycosides	J01GB12	Watch	No	0
Aspoxicillin	Penicillins	J01CA19	Watch	No	0
Azithromycin	Macrolides	J01FA10	Watch	Yes	0
Azlocillin	Penicillins	J01CA09	Watch	No	0
Bekanamycin	Aminoglycosides	J01GB13	Watch	No	0
Biapenem	Carbapenems	J01DH05	Watch	No	0
Carbenicillin	Penicillins	J01CA03	Watch	No	0
Carindacillin	Penicillins	J01CA05	Watch	No	0
Cefaclor	Second-generation-cephalosporins	J01DC04	Watch	No	0
Cefamandole	Second-generation-cephalosporins	J01DC03	Watch	No	0
Cefbuperazone	Second-generation-cephalosporins	J01DC13	Watch	No	0
Cefcapene-pivoxil	Third-generation-cephalosporins	J01DD17	Watch	No	0
Cefdinir	Third-generation-cephalosporins	J01DD15	Watch	No	0
Cefditoren-pivoxil	Third-generation-cephalosporins	J01DD16	Watch	No	0
Cefepime	Fourth-generation-cephalosporins	J01DE01	Watch	No	0
Cefetamet-pivoxil	Third-generation-cephalosporins	J01DD10	Watch	No	0
Cefixime	Third-generation-cephalosporins	J01DD08	Watch	Yes	0
Cefmenoxime	Third-generation-cephalosporins	J01DD05	Watch	No	0
Cefmetazole	Second-generation-cephalosporins	J01DC09	Watch	No	0
Cefminox	Second-generation-cephalosporins	J01DC12	Watch	No	0
Cefodizime	Third-generation-cephalosporins	J01DD09	Watch	No	0
Cefonicid	Second-generation-cephalosporins	J01DC06	Watch	No	0
Cefoperazone	Third-generation-cephalosporins	J01DD12	Watch	No	0
Ceforanide	Second-generation-cephalosporins	J01DC11	Watch	No	0
Cefoselis	Fourth-generation-cephalosporins	not assigned	Watch	No	0
Cefotaxime	Third-generation-cephalosporins	J01DD01	Watch	Yes	0
Cefotetan	Second-generation-cephalosporins	J01DC05	Watch	No	0
Cefotiam	Second-generation-cephalosporins	J01DC07	Watch	No	0
Cefoxitin	Second-generation-cephalosporins	J01DC01	Watch	No	0
Cefozopran	Fourth-generation-cephalosporins	J01DE03	Watch	No	0
Cefpiramide	Third-generation-cephalosporins	J01DD11	Watch	No	0
Cefpirome	Fourth-generation-cephalosporins	J01DE02	Watch	No	0
Cefpodoxime-proxetil	Third-generation-cephalosporins	J01DD13	Watch	No	0
Cefprozil	Second-generation-cephalosporins	J01DC10	Watch	No	0
Cefsulodin	Third-generation-cephalosporins	J01DD03	Watch	No	0
Ceftazidime	Third-generation-cephalosporins	J01DD02	Watch	Yes	0
Cefteram-pivoxil	Third-generation-cephalosporins	J01DD18	Watch	No	0
Ceftibuten	Third-generation-cephalosporins	J01DD14	Watch	No	0
Ceftizoxime	Third-generation-cephalosporins	J01DD07	Watch	No	0
Ceftriaxone	Third-generation-cephalosporins	J01DD04	Watch	Yes	0
Cefuroxime	Second-generation-cephalosporins	J01DC02	Watch	Yes	0
Chlortetracycline	Tetracyclines	J01AA03	Watch	No	0
Cinoxacin	Quinolones	J01MB06	Watch	No	0
Ciprofloxacin	Fluoroquinolones	J01MA02	Watch	Yes	0
Clarithromycin	Macrolides	J01FA09	Watch	Yes	0
Clofoctol	Phenol-derivatives	J01XX03	Watch	No	0
Clomocycline	Tetracyclines	J01AA11	Watch	No	0
Delafoxacin	Fluoroquinolones	J01MA23	Watch	No	0
Demeclocycline	Tetracyclines	J01AA01	Watch	No	0
Dibekacin	Aminoglycosides	J01GB09	Watch	No	0
Dirithromycin	Macrolides	J01FA13	Watch	No	0
Doripenem	Carbapenems	J01DH04	Watch	No	0
Enoxacin	Fluoroquinolones	J01MA04	Watch	No	0
Ertapenem	Carbapenems	J01DH03	Watch	No	0
Erythromycin	Macrolides	J01FA01	Watch	Yes	*therapeutic alternative to clarithromycin (EMLc)
Fidaxomicin	Macrolides	A07AA12	Watch	No	0
Fleroxacin	Fluoroquinolones	J01MA08	Watch	No	0
Flomoxef	Second-generation-cephalosporins	J01DC14	Watch	No	0
Flumequine	Quinolones	J01MB07	Watch	No	0
Flurithromycin	Macrolides	J01FA14	Watch	No	0
Fosfomycin_oral	Phosphonics	J01XX01	Watch	No	0
Fusidic-acid	Steroid-antibacterials	J01XC01	Watch	No	0
Garenoxacin	Fluoroquinolones	J01MA19	Watch	No	0
Gatifloxacin	Fluoroquinolones	J01MA16	Watch	No	0
Gemifloxacin	Fluoroquinolones	J01MA15	Watch	No	0
Gepotidacin	Triazaacenaphthylenes	J01XX13	Watch	No	0
Grepafloxacin	Fluoroquinolones	J01MA11	Watch	No	0
Imipenem/cilastatin	Carbapenems	J01DH51	Watch	Yes	*therapeutic alternative to meropenem
Isepamicin	Aminoglycosides	J01GB11	Watch	No	0
Josamycin	Macrolides	J01FA07	Watch	No	0
Kanamycin_IV	Aminoglycosides	J01GB04	Watch	No	0
Kanamycin_oral	Aminoglycosides	A07AA08	Watch	No	0
Lascufloxacin	Fluoroquinolones	J01MA25	Watch	No	0
Latamoxef	Third-generation-cephalosporins	J01DD06	Watch	No	0
Levofloxacin	Fluoroquinolones	J01MA12	Watch	No	0
Levonadifloxacin	Fluoroquinolones	J01MA24	Watch	No	0
Lincosamycin	Lincosamides	J01FF02	Watch	No	0
Lomefloxacin	Fluoroquinolones	J01MA07	Watch	No	0
Loracarbef	Second-generation-cephalosporins	J01DC08	Watch	No	0
Lymecycline	Tetracyclines	J01AA04	Watch	No	0
Meropenem	Carbapenems	J01DH02	Watch	Yes	0
Metacycline	Tetracyclines	J01AA05	Watch	No	0
Mezlocillin	Penicillins	J01CA10	Watch	No	0
Micronomicin	Aminoglycosides	not assigned	Watch	No	0
Midecamycin	Macrolides	J01FA03	Watch	No	0
Minocycline_oral	Tetracyclines	J01AA08	Watch	No	0
Miocamycin	Macrolides	J01FA11	Watch	No	0
Moxifloxacin	Fluoroquinolones	J01MA14	Watch	No	0
Nalidixic-acid	Quinolones	J01MB02	Watch	No	0
Nemonoxacin	Quinolones	J01MB08	Watch	No	0
Neomycin_IV	Aminoglycosides	J01GB05	Watch	No	0

Neomycin_oral	Aminoglycosides	A07AA01	Watch	No	0
Netilmicin	Aminoglycosides	J01GB07	Watch	No	0
Norfloxacin	Fluoroquinolones	J01MA06	Watch	No	0
Ofloxacin	Fluoroquinolones	J01MA01	Watch	No	0
Oleandomycin	Macrolides	J01FA05	Watch	No	0
Oxolinic-acid	Quinolones	J01MB05	Watch	No	0
Oxytetracycline	Tetracyclines	J01AA06	Watch	No	0
Panipenem	Carbapenems	J01DH55	Watch	No	0
Paromomycin_oral	Aminoglycosides	A07AA06	Watch	No	0
Pazufloxacin	Fluoroquinolones	J01MA18	Watch	No	0
Pefloxacin	Fluoroquinolones	J01MA03	Watch	No	0
Penimepicycline	Tetracyclines	J01AA10	Watch	No	0
Pheneticillin	Penicillins	J01CE05	Watch	No	0
Pipemidic-acid	Quinolones	J01MB04	Watch	No	0
Piperacillin	Penicillins	J01CA12	Watch	No	0
Piperacillin/tazobactam	Beta-lactam/beta-lactamase-inhibitor_anti-ps	J01CR05	Watch	Yes	0
Piromidic-acid	Quinolones	J01MB03	Watch	No	0
Pristinamycin	Streptogramins	J01FG01	Watch	No	0
Prulifloxacin	Fluoroquinolones	J01MA17	Watch	No	0
Ribostamycin	Aminoglycosides	J01GB10	Watch	No	0
Rifabutin	Rifamycins	J04AB04	Watch	No	0
Rifampicin	Rifamycins	J04AB02	Watch	No	0
Rifamycin_IV	Rifamycins	J04AB03	Watch	No	0
Rifamycin_oral	Rifamycins	A07AA13	Watch	No	0
Rifaximin	Rifamycins	A07AA11	Watch	No	0
Rokitamycin	Macrolides	J01FA12	Watch	No	0
Rolitetracycline	Tetracyclines	J01AA09	Watch	No	0
Rosoxacin	Quinolones	J01MB01	Watch	No	0
Roxithromycin	Macrolides	J01FA06	Watch	No	0
Rufloxacin	Fluoroquinolones	J01MA10	Watch	No	0
Sarecycline	Tetracyclines	J01AA14	Watch	No	0
Sisomicin	Aminoglycosides	J01GB08	Watch	No	0
Sitafloxacin	Fluoroquinolones	J01MA21	Watch	No	0
Solithromycin	Macrolides	J01FA16	Watch	No	0
Sparfloxacin	Fluoroquinolones	J01MA09	Watch	No	0
Spiramycin	Macrolides	J01FA02	Watch	No	0
Streptoduocin	Aminoglycosides	J01GA02	Watch	No	0
Streptomycin_IV	Aminoglycosides	J01GA01	Watch	No	0
Streptomycin_oral	Aminoglycosides	A07AA04	Watch	No	0
Sulbenicillin	Penicillins	J01CA16	Watch	No	0
Tazobactam	Beta-lactamase-inhibitors	J01CG02	Watch	No	0
Tebipenem	Carbapenems	J01DH06	Watch	No	0
Teicoplanin	Glycopeptides	J01XA02	Watch	No	0
Telithromycin	Macrolides	J01FA15	Watch	No	0
Temaflloxacin	Fluoroquinolones	J01MA05	Watch	No	0
Temocillin	Penicillins	J01CA17	Watch	No	0
Ticarcillin	Penicillins	J01CA13	Watch	No	0
Ticarcillin/clavulanic-acid	Beta-lactam/beta-lactamase-inhibitor	J01CR03	Watch	No	0
Tobramycin	Aminoglycosides	J01GB01	Watch	No	0
Tosufloxacin	Fluoroquinolones	J01MA22	Watch	No	0
Troleandomycin	Macrolides	J01FA08	Watch	No	0
Trovaflloxacin	Fluoroquinolones	J01MA13	Watch	No	0
Vancomycin_IV	Glycopeptides	J01XA01	Watch	Yes	0
Vancomycin_oral	Glycopeptides	A07AA09	Watch	Yes	0

Reserve group antibiotics

This group includes antibiotics and antibiotic classes that should be reserved for treatment of confirmed or suspected infections due to multi-drug-resistant organisms. Reserve group antibiotics should be treated as "last resort" options.

Selected Reserve group antibiotics are listed as individual medicines on the WHO Model Lists of Essential Medicines when they have a favourable risk-benefit profile and proven activity against "Critical Priority" or "High Priority" pathogens identified by the WHO Priority Pathogens List¹, notably carbapenem resistant Enterobacteriaceae. These antibiotics should be accessible, but their use should be tailored to highly specific patients and settings, when all alternatives have failed or are not suitable.

These medicines could be protected and prioritized as key targets of national and international stewardship programs involving monitoring and utilization reporting, to preserve their effectiveness.

¹ WHO Priority Pathogens List

Antibiotic	Class	ATC code	Category	Listed on EML/EMLc 2025	Notes
Aztreonam	Monobactams	J01DF01	Reserve	No	0
Aztreonam/avibactam	Monobactam/beta-lactamase-inhibitor	J01DF51	Reserve	No	0
Carumonam	Monobactams	J01DF02	Reserve	No	0
Cefiderocol	Other-cephalosporins	J01DI04	Reserve	Yes	0
Ceftaroline-fosamil	Fifth-generation-cephalosporins	J01DI02	Reserve	No	0
Ceftazidime/avibactam	Third-generation-cephalosporins	J01DD52	Reserve	Yes	0
Ceftobiprole-medocartil	Fifth-generation-cephalosporins	J01DI01	Reserve	No	0
Ceftolozane/tazobactam	Fifth-generation-cephalosporins	J01DI54	Reserve	Yes	0
Colistin_IV	Polymyxins	J01XB01	Reserve	Yes	0
Colistin_oral	Polymyxins	A07AA10	Reserve	No	0
Dalbavancin	Glycopeptides	J01XA04	Reserve	No	0
Dalfopristin/quinupristin	Streptogramins	J01FG02	Reserve	No	0
Daptomycin	Lipopeptides	J01XX09	Reserve	No	0
Eravacycline	Tetracyclines	J01AA13	Reserve	No	0
Faropenem	Penems	J01DI03	Reserve	No	0
Fosfomycin_IV	Phosphonics	J01XX01	Reserve	Yes	0
Iclaprim	Trimethoprim-derivatives	J01EA03	Reserve	No	0
Imipenem/cilastatin/relebactam	Carbapenems	J01DH56	Reserve	No	0
Lefamulin	Pleuromutilin	J01XX12	Reserve	No	0
Linezolid	Oxazolidinones	J01XX08	Reserve	Yes	0
Meropenem/vaborbactam	Carbapenems	J01DH52	Reserve	Yes	0
Minocycline_IV	Tetracyclines	J01AA08	Reserve	No	0
Omadacycline	Tetracyclines	J01AA15	Reserve	No	0
Oritavancin	Glycopeptides	J01XA05	Reserve	No	0
Plazomicin	Aminoglycosides	J01GB14	Reserve	Yes	0
Polymyxin-B_IV	Polymyxins	J01XB02	Reserve	Yes	0
Polymyxin-B_oral	Polymyxins	A07AA05	Reserve	No	0
Tedizolid	Oxazolidinones	J01XX11	Reserve	Yes	*therapeutic alternative to linezolid (EML)
Telavancin	Glycopeptides	J01XA03	Reserve	No	0
Tigecycline	Glycylcyclines	J01AA12	Reserve	No	0

Antibiotics not recommended

The use of the fixed-dose combinations of multiple broad-spectrum antibiotics listed here is not evidence-based, nor recommended in high-quality international guidelines. WHO does not recommend their use in clinical practice.

Antibiotic

acetylspiramycin/metronidazole
amikacin/cefepime
amoxicillin/bacillus coagulans/cloxacillin
amoxicillin/bacillus coagulans/dicloxacillin
amoxicillin/clavulanic acid/lactic ferments
amoxicillin/clavulanic acid/lactobacillus acidophilus
amoxicillin/clavulanic acid/nimesulide
amoxicillin/cloxacillin
amoxicillin/cloxacillin/lactic acid
amoxicillin/cloxacillin/lactobacillus acidophilus/serrapeptase
amoxicillin/cloxacillin/lactobacillus lactis
amoxicillin/cloxacillin/serrapeptase
amoxicillin/dicloxacillin
amoxicillin/dicloxacillin/saccharomyces boulardii
amoxicillin/flucloxacillin
amoxicillin/flucloxacillin/lactobacillus acidophilus
amoxicillin/metronidazole
amoxicillin/pivsulbactam
ampicillin/bacillus coagulans/cloxacillin
ampicillin/cloxacillin
ampicillin/cloxacillin/lactobacillus acidophilus
ampicillin/cloxacillin/saccharomyces boulardii
ampicillin/dicloxacillin
ampicillin/dicloxacillin/lactobacillus acidophilus
ampicillin/flucloxacillin
ampicillin/lidocaine/sulbactam
ampicillin/oxacillin
ampicillin/sultamicillin
ascorbic acid/metamizole sodium/penicillin g /streptomycin
azithromycin/cefixime
azithromycin/cefixime/lactobacillus acidophilus
azithromycin/cefpodoxime proxetil
azithromycin/fluconazole/secnidazole
azithromycin/levofloxacin
azithromycin/ofloxacin
benzyl penicillin/streptomycin
benzylpenicillin/procaine-benzylpenicillin/benzathine-benzylpenicillin
bromelains/doxycycline/lactobacillus reuteri/lactobacillus rhamnosus/ornidazole
bromhexine/sulfamethoxazole/trimethoprim
cefaclor/clavulanic acid
cefadroxil/clavulanic acid
cefadroxil/trimethoprim
cefaletin/trimethoprim
cefdinir/clavulanic acid
cefepime/sulbactam
cefepime/tazobactam
cefixime/cefpodoxime proxetil
cefixime/clavulanic acid
cefixime/clavulanic acid/lactobacillus acidophilus
cefixime/cloxacillin
cefixime/cloxacillin/lactobacillus acidophilus
cefixime/dicloxacillin
cefixime/lactobacillus acidophilus/ofloxacin
cefixime/levofloxacin
cefixime/linezolid
cefixime/moxifloxacin
cefixime/ofloxacin
cefixime/ornidazole
cefoperazone/sulbactam
cefoperazone/tazobactam
cefotaxime/sulbactam
cefpodoxime proxetil/clavulanic acid
cefpodoxime proxetil/cloxacillin/lactobacillus acidophilus
cefpodoxime proxetil/dicloxacillin
cefpodoxime proxetil/dicloxacillin/lactobacillus acidophilus
cefpodoxime proxetil/levofloxacin
cefpodoxime proxetil/ofloxacin
cefpodoxime proxetil/sulbactam
ceftazidime/sulbactam
ceftazidime/tazobactam
ceftazidime/tobramicin
ceftibuten/clavulanic acid
ceftriaxone/sulbactam
ceftriaxone/tazobactam
ceftriaxone/vancomycin
cefuroxime axetil/clavulanic acid
cefuroxime axetil/linezolid
cefuroxime axetil/sulbactam
cefuroxime/clavulanic acid
cefuroxime/metronidazole

cefuroxime/sulbactam
chloramphenicol/tetracycline
ciprofloxacin/metronidazole
ciprofloxacin/ornidazole
ciprofloxacin/tinidazole
doxycycline/tinidazole
erythromycin/sulfamethoxazole/trimethoprim
erythromycin/trimethoprim
fosfomycin/trimethoprim
gatifloxacin/ornidazole
kanamycin/penicillin g
levofloxacin/metronidazole
levofloxacin/ornidazole
meropenem/sodium/sulbactam
meropenem/sulbactam
metronidazole/diloxanide
metronidazole/furazolidone
metronidazole/norfloxacin
metronidazole/spiramycin
metronidazole/tetracycline
mezlocillin/sulbactam
norfloxacin/tinidazole
ofloxacin/nitazoxanide
ofloxacin/ornidazole
ofloxacin/tinidazole
oleandomycin/tetracycline
penicillins/combinations-with-other-antibacterials
piperacillin/sulbactam
pivampicillin/pivmecillinam
rifampicin/trimethoprim
sulfacarbamide/sulfadiazine/sulfadimidine
sulfadiazine/sulfamethoxazole/trimethoprim
sulfonamides/combinations-with-other-antibacterials- (excl-trimethoprim)
tetracycline/chlortetracycline/demeclocycline
tetracycline/nystatin
tinidazole/diloxanide

Antibiotics on 24th WHO Model List of Essential Medicines 2025

Section 6.2.1 Access group antibiotics

Section 6.2.2 Watch group antibiotics

Section 6.2.3 Reserve group antibiotics

Antibiotic	Class	ATC code	Category	EML 2025	Notes
Amikacin	Aminoglycosides	J01GB06	Access	Yes	0
Amoxicillin	Penicillins	J01CA04	Access	Yes	0
Amoxicillin/clavulanic-acid	Beta-lactam/beta-lactamase-inhibitor	J01CR02	Access	Yes	0
Ampicillin	Penicillins	J01CA01	Access	Yes	0
Azithromycin	Macrolides	J01FA10	Watch	Yes	0
Benzathine-benzylpenicillin	Penicillins	J01CE08	Access	Yes	0
Benzylpenicillin	Penicillins	J01CE01	Access	Yes	0
Cefalexin	First-generation-cephalosporins	J01DB01	Access	Yes	0
Cefazolin	First-generation-cephalosporins	J01DB04	Access	Yes	0
Cefiderocol	Other-cephalosporins	J01DI04	Reserve	Yes	0
Cefixime	Third-generation-cephalosporins	J01DD08	Watch	Yes	0
Cefotaxime	Third-generation-cephalosporins	J01DD01	Watch	Yes	0
Ceftazidime	Third-generation-cephalosporins	J01DD02	Watch	Yes	0
Ceftazidime/avibactam	Third-generation-cephalosporins	J01DD52	Reserve	Yes	0
Ceftolozane/tazobactam	Fifth-generation-cephalosporins	J01DI54	Reserve	Yes	0
Ceftriaxone	Third-generation-cephalosporins	J01DD04	Watch	Yes	0
Cefuroxime	Second-generation-cephalosporins	J01DC02	Watch	Yes	0
Chloramphenicol	Amphenicols	J01BA01	Access	Yes	0
Ciprofloxacin	Fluoroquinolones	J01MA02	Watch	Yes	0
Clarithromycin	Macrolides	J01FA09	Watch	Yes	0
Clindamycin	Lincosamides	J01FF01	Access	Yes	0
Cloxacillin	Penicillins	J01CF02	Access	Yes	0
Colistin_IV	Polymyxins	J01XB01	Reserve	Yes	0
Dicloxacillin	Penicillins	J01CF01	Access	Yes	*therapeutic alternative to cloxacillin
Doxycycline	Tetracyclines	J01AA02	Access	Yes	0
Erythromycin	Macrolides	J01FA01	Watch	Yes	*therapeutic alternative to clarithromycin (EMLc)
Flucloxacillin	Penicillins	J01CF05	Access	Yes	*therapeutic alternative to cloxacillin
Fosfomycin_IV	Phosphonics	J01XX01	Reserve	Yes	0
Gentamicin	Aminoglycosides	J01GB03	Access	Yes	0
Imipenem/cilastatin	Carbapenems	J01DH51	Watch	Yes	*therapeutic alternative to meropenem
Linezolid	Oxazolidinones	J01XX08	Reserve	Yes	0
Meropenem	Carbapenems	J01DH02	Watch	Yes	0
Meropenem/vaborbactam	Carbapenems	J01DH52	Reserve	Yes	0
Meticillin	Penicillins	J01CF03	Access	Yes	*therapeutic alternative to cloxacillin
Metronidazole_IV	Imidazoles	J01XD01	Access	Yes	0
Metronidazole_oral	Imidazoles	P01AB01	Access	Yes	0
Nafcillin	Penicillins	J01CF06	Access	Yes	*therapeutic alternative to cloxacillin
Nitrofurantoin	Nitrofurans-derivatives	J01XE01	Access	Yes	0
Oxacillin	Penicillins	J01CF04	Access	Yes	*therapeutic alternative to cloxacillin
Phenoxymethylpenicillin	Penicillins	J01CE02	Access	Yes	0
Piperacillin/tazobactam	Beta-lactam/beta-lactamase-inhibitor_anti-ps	J01CR05	Watch	Yes	0
Plazomicin	Aminoglycosides	J01GB14	Reserve	Yes	0
Polymyxin-B_IV	Polymyxins	J01XB02	Reserve	Yes	0
Procaine-benzylpenicillin	Penicillins	J01CE09	Access	Yes	0
Spectinomycin	Aminocyclitols	J01XX04	Access	Yes	0
Sulfamethoxazole/trimethoprim	Sulfonamide-trimethoprim-combinations	J01EE01	Access	Yes	0
Tedizolid	Oxazolidinones	J01XX11	Reserve	Yes	*therapeutic alternative to linezolid (EML)
Trimethoprim	Trimethoprim-derivatives	J01EA01	Access	Yes	0
Vancomycin_IV	Glycopeptides	J01XA01	Watch	Yes	0
Vancomycin_oral	Glycopeptides	A07AA09	Watch	Yes	0

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Section 6.2.1 Access group antibiotics

Section 6.2.2 Watch group antibiotics

Section 6.2.3 Reserve group antibiotics

Antibiotic	Class	ATC code	Category	EMLc 2023	Notes
Amikacin	Aminoglycosides	J01GB06	Access	Yes	0
Amoxicillin	Penicillins	J01CA04	Access	Yes	0
Amoxicillin/clavulanic-acid	Beta-lactam/beta-lactamase-inhibitor	J01CR02	Access	Yes	0
Ampicillin	Penicillins	J01CA01	Access	Yes	0
Azithromycin	Macrolides	J01FA10	Watch	Yes	0
Benzathine-benzylpenicillin	Penicillins	J01CE08	Access	Yes	0
Benzylpenicillin	Penicillins	J01CE01	Access	Yes	0
Cefalexin	First-generation-cephalosporins	J01DB01	Access	Yes	0
Cefazolin	First-generation-cephalosporins	J01DB04	Access	Yes	0
Cefiderocol	Other-cephalosporins	J01DI04	Reserve	Yes	0
Cefixime	Third-generation-cephalosporins	J01DD08	Watch	Yes	0
Cefotaxime	Third-generation-cephalosporins	J01DD01	Watch	Yes	0
Ceftazidime	Third-generation-cephalosporins	J01DD02	Watch	Yes	0
Ceftazidime/avibactam	Third-generation-cephalosporins	J01DD52	Reserve	Yes	0
Ceftolozane/tazobactam	Fifth-generation-cephalosporins	J01DI54	Reserve	Yes	0
Ceftriaxone	Third-generation-cephalosporins	J01DD04	Watch	Yes	0
Cefuroxime	Second-generation-cephalosporins	J01DC02	Watch	Yes	0
Chloramphenicol	Amphenicols	J01BA01	Access	Yes	0
Ciprofloxacin	Fluoroquinolones	J01MA02	Watch	Yes	0
Clarithromycin	Macrolides	J01FA09	Watch	Yes	0
Clindamycin	Lincosamides	J01FF01	Access	Yes	0
Cloxacillin	Penicillins	J01CF02	Access	Yes	0
Colistin_IV	Polymyxins	J01XB01	Reserve	Yes	0
Dicloxacillin	Penicillins	J01CF01	Access	Yes	*therapeutic alternative to cloxacillin
Doxycycline	Tetracyclines	J01AA02	Access	Yes	0
Erythromycin	Macrolides	J01FA01	Watch	Yes	*therapeutic alternative to clarithromycin (EMLc)
Flucloxacillin	Penicillins	J01CF05	Access	Yes	*therapeutic alternative to cloxacillin
Fosfomycin_IV	Phosphonics	J01XX01	Reserve	Yes	0
Gentamicin	Aminoglycosides	J01GB03	Access	Yes	0
Imipenem/cilastatin	Carbapenems	J01DH51	Watch	Yes	*therapeutic alternative to meropenem
Linezolid	Oxazolidinones	J01XX08	Reserve	Yes	0
Meropenem	Carbapenems	J01DH02	Watch	Yes	0
Meropenem/vaborbactam	Carbapenems	J01DH52	Reserve	Yes	0
Meticillin	Penicillins	J01CF03	Access	Yes	*therapeutic alternative to cloxacillin
Metronidazole_IV	Imidazoles	J01XD01	Access	Yes	0
Metronidazole_oral	Imidazoles	P01AB01	Access	Yes	0
Nafcillin	Penicillins	J01CF06	Access	Yes	*therapeutic alternative to cloxacillin
Nitrofurantoin	Nitrofurans-derivatives	J01XE01	Access	Yes	0
Oxacillin	Penicillins	J01CF04	Access	Yes	*therapeutic alternative to cloxacillin
Phenoxymethylpenicillin	Penicillins	J01CE02	Access	Yes	0
Piperacillin/tazobactam	Beta-lactam/beta-lactamase-inhibitor_anti-ps	J01CR05	Watch	Yes	0
Plazomicin	Aminoglycosides	J01GB14	Reserve	Yes	0
Polymyxin-B_IV	Polymyxins	J01XB02	Reserve	Yes	0
Procaine-benzylpenicillin	Penicillins	J01CE09	Access	Yes	0
Spectinomycin	Aminocyclitols	J01XX04	Access	Yes	0
Sulfamethoxazole/trimethoprim	Sulfonamide-trimethoprim-combinations	J01EE01	Access	Yes	0
Tedizolid	Oxazolidinones	J01XX11	Reserve	Yes	*therapeutic alternative to linezolid (EML)
Trimethoprim	Trimethoprim-derivatives	J01EA01	Access	Yes	0
Vancomycin_IV	Glycopeptides	J01XA01	Watch	Yes	0
Vancomycin_oral	Glycopeptides	A07AA09	Watch	Yes	0